

Habitat Relations

Selection of Landscapes by Male Ruffed Grouse During Peak Abundance

MEADOW J. KOUFFELD,¹ *Department of Fisheries, Wildlife, and Conservation Biology, University of Minnesota, Saint Paul, MN 55108, USA*
 MICHAEL A. LARSON, *Forest Wildlife Populations and Research Group, Minnesota Department of Natural Resources, Grand Rapids, MN 55744, USA*
 R. J. GUTIÉRREZ,² *Department of Fisheries, Wildlife, and Conservation Biology, University of Minnesota, Saint Paul, MN 55108, USA*

ABSTRACT We studied the habitat relationships of male ruffed grouse (*Bonasa umbellus*) at the landscape scale when grouse were at the peak of their population cycle. We located 290 and 230 male grouse during 2009 and 2010, respectively, on 30 landscapes encompassing 5,349 ha. We used an information-theoretic model selection approach to examine 2 sets of a priori models. We designed our first model set to evaluate the relationships between Shannon's evenness index of cover types and grouse density and our second set to evaluate other relationships between grouse density and landscape configuration. The top model in the first set was a null model (intercepts only), but a competing model indicated that Shannon's evenness index was positively correlated with male grouse density ($r = 0.43$). Shannon's evenness index was confounded with cover type dominance in landscapes because it was positively correlated ($r = 0.55$) with the proportion of the aspen cover type and negatively correlated ($r = -0.79$) with the proportion of the conifer cover type. The top-ranked model from our second model set included only year and road density. Road density was negatively related to grouse density ($r = -0.34$), which could mean either hunting pressure along roads affected density or cover types were different in landscapes with greater road densities. The year-only model indicated that male grouse density was less in 2010 than in 2009 ($\beta_{\text{year}} = -0.014$, 95% CI = -0.024 to -0.005). We predict that forest management practices that result in an even distribution of cover types within landscapes should result in greater densities of grouse. © 2013 The Wildlife Society.

KEY WORDS *Bonasa umbellus*, density, habitat selection, landscape, road density, ruffed grouse, Shannon's evenness index.

The ruffed grouse (*Bonasa umbellus*) is widely distributed in North America (Rusch et al. 2000). It is closely associated with early successional habitats, particularly aspen forest (*Populus* spp.; Rusch et al. 2000). Ruffed grouse are hunted throughout their range; hence, their population dynamics and habitat selection are of interest to wildlife managers (Rusch et al. 2000). Although ruffed grouse occur in many habitats, they reach their greatest density in areas with a mosaic of different-aged aspen stands in close proximity (Gullion and Alm 1983, Rusch et al. 2000). This suggests that aspen-dominated landscapes may confer higher fitness to grouse relative to other habitats (Gullion and Marshall 1968, Zimmerman et al. 2009).

In contrast, conifers might have either negative or beneficial effects on grouse, depending on the circumstances (Bump et al. 1947). For example, conifers may provide hunting perches and nesting areas for avian predators (Gullion and Marshall 1968, Gullion and Alm 1983), whereas they may provide thermal cover and cover from

predators when snow is not available for roosting (Thompson and Fritzell 1988).

Most habitat research on ruffed grouse has focused on small spatial scales (i.e., male activity centers or forest stands; e.g., Gullion and Alm 1983). However, Zimmerman et al. (2009) examined the influence of landscape-scale characteristics on grouse density. They found the greatest density of grouse in landscapes with the greatest Shannon's evenness index values, but their finding was confounded because landscapes that had greater Shannon's evenness index values also had a greater proportion of aspen cover and a lesser proportion of conifer cover (Zimmerman et al. 2009).

The resolution of this confounding issue is needed for 2 reasons: 1) proposals have been made to shift landscape composition toward pre-settlement conditions by increasing the abundance of conifer forests, and 2) some forest industries prefer to grow conifers for economic reasons (Minnesota Department of Natural Resources [MNDNR] 2009). Thus, cover type conversions from aspen to conifer may reduce grouse densities, which would negatively affect ruffed grouse hunting. Therefore, we evaluated whether the results of Zimmerman et al. (2009) were applicable to other landscapes because their study area (Cloquet Forestry Center) was atypical of natural forests (i.e., cover types were highly heterogeneous because of a history of forestry

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¹Present address: Minnesota Department of Natural Resources, International Falls, MN 56649

²E-mail: gutie012@umn.edu

experiments). In addition, we used the Zimmerman et al. (2009) study as a basis to evaluate other potential landscape-scale effects they could not consider because of their unique study area.

STUDY AREA

We studied male ruffed grouse during 2009–2010 on the Red Lake Wildlife Management Area and Beltrami Island State Forest in north central Minnesota (see Fig. S1 in supplemental methods Appendix S1, available online at www.onlinelibrary.wiley.com). Our 2 adjacent study areas encompassed approximately 251,038 ha and were located in the Laurentian Mixed Forest Province (Bailey 1995). This floristic province encompassed a major portion of ruffed grouse range in North America, extending from north central Minnesota to southern Canada and east to New England. This province was characterized by conifer forests, hardwood forests, mixed hardwood and conifer forests, conifer bogs, and marshes. Annual precipitation in our study areas was 55.8 cm in 2009 and 84.9 cm in 2010 (G. Mehmehl, MNDNR, unpublished data). Winters were cold (mean daily low temperature was -26.8°C and -18°C during January 2009 and 2010, respectively) and summers were warm (mean daily high temperature was 22°C and 26°C during July 2009 and 2010, respectively). Topography was relatively flat with elevations ranging from 369 m to 410 m above sea level. Approximately 33.9% of our study areas were upland cover types (forested and un-forested uplands) and the remainder was lowland cover types (forested and un-forested wetlands).

Our northern area (116,454 ha) was dominated by conifer forests and our southern area (134,584 ha) was dominated by aspen forests. Lowland areas in the north were forests comprised of black spruce (*Picea mariana*), tamarack (*Larix laricina*), speckled alder (*Alnus incana* spp. *rugosa*), and eastern white-cedar (*Thuja occidentalis*); upland areas were primarily mixed or pure forests dominated by red pine (*Pinus resinosa*), aspen (primarily *Populus tremuloides* and secondarily by *P. grandidentata*), balsam poplar (*Populus balsamifera*), jack pine (*Pinus banksiana*), balsam fir (*Abies balsamea*), birch (*Betula papyrifera*, *B. alleghaniensis*), white pine (*Pinus strobus*), and hardwoods such as green ash (*Fraxinus pennsylvanica*), black ash (*F. nigra*), and oak (*Quercus* spp.). The southern area lowlands were dominated by ash species and tamarack; upland areas were dominated by forests of either pure or mixed aspen, balsam poplar, red pine, white pine, white spruce, and balsam fir. The understory of both study areas was dominated by a variety of shrubs such as hazel (*Corylus* spp.), mountain maple (*Acer spicatum*), red-osier dogwood (*Cornus sericea*), and speckled alder.

METHODS

Data Collection

We selected 60 random points (30 within each study area) and then located the nearest road or trail to this point to serve as the starting point of a sampling transect. We then randomly selected the direction of transects along these roads

or trails. We were restricted to roads and trails because much of the study area was covered with sphagnum bogs or swamps, many of which were not passable during the spring and summer. Moreover, because of logging and trails maintained for recreation, any male grouse on our study areas could be potentially heard displaying from a road or trail. Each transect was 3–5 km in length (see Kouffeld 2011 for details). We divided our 60 transects into 3 groups that represented 3 categories of cover type dominance: 1) conifer dominated (hereafter conifer), 2) aspen-birch dominated (hereafter aspen), and 3) mixed (proportion of aspen and conifer cover types was similar). We then randomly selected 10 transects from each of the 3 groups, regardless of study area. We considered the area of these transects to be our landscapes for purposes of analysis (see below).

We surveyed for displaying (drumming) males using methods described by Zimmerman and Gutiérrez (2007). We only included grouse detected within 200 m of our transect line because detection probability for grouse beyond 200 m is low (Zimmerman and Gutiérrez 2007). We conducted 1 drumming survey per transect per week for 8 weeks on 30 transects (i.e., landscapes) during each breeding season (6 April to 29 May 2009 and 5 April to 28 May 2010). This time period represented the primary breeding season of grouse in northern Minnesota (Zimmerman 2006). Individual landscapes ranged from 121 ha to 156 ha and collectively encompassed 5,349 ha. We surveyed on foot starting 0.5 hr before sunrise and continued until the survey (transect) was completed (approx. 2.5–6 hr). We reversed travel direction on each subsequent survey because drumming activity wanes after sunrise (Rusch et al. 2000, Zimmerman and Gutiérrez 2007). We determined the location of displaying grouse we detected by first triangulating on the drumming sound and then stealthily approaching the displaying bird to observe each grouse on its drumming structure. If a bird flushed before visual location, we searched the approximate area of the flush to identify likely drumming structures (i.e., accumulation of fecal droppings on or next to a structure; Gullion and Marshall 1968). We recorded the location of drumming structures using a handheld global positioning system (GPS; Garmin GPSmap 76S, Garmin Ltd., Olathe, KS) and then marked them with flagging and a numbered metal tag nailed to the structure. We avoided double counting grouse by distinguishing between their primary and secondary drumming structures and observing birds on their primary drumming stages on multiple occasions (Rusch et al. 2000, Zimmerman and Gutiérrez 2007; see Tables S1 and S2 in supplemental methods Appendix S1, available online at www.onlinelibrary.wiley.com). Our research protocol was approved by the University of Minnesota IACUC (protocol # 0912A75021).

Our transect width for surveys was 400 m; we only used birds detected within 175 m of the transect center (350 m total transect width) to minimize truncation of the size of the cover type actually occupied by a grouse when we measured landscape characteristics. Assuming a detection probability of 0.31 for displaying grouse (Zimmerman and Gutiérrez 2007), the

probability we detected a drumming grouse within this distance at least once during our 8 surveys was approximately 0.95.

We derived 3 different cover type classifications using MNDNR Forest Stand Inventory data (MNDNR 2010) and United States Geological Survey digital orthophoto quadrangles as a basemap for different analyses; positional accuracy was that of the standard 24k digital raster graphics. Each classification included 2 non-forest cover types and a forest cover type: 1) 6 cover types based on tree species composition (Zimmerman et al. 2009; see also Table S3 in supplemental methods Appendix S1, available online at www.onlinelibrary.wiley.com), 2) 9 cover types based on cover type associations of grouse, their predators, or cover type physiognomy without regard to age of forest stands (Gullion and Alm 1983), and 3) 21 cover type associations of grouse, their predators, or cover type physiognomy including forest age class. To be consistent with Zimmerman's (2006) method of classification, we assigned stands with <66% basal area dominance by a particular tree species or age class to a mixed forest cover type. We used existing land cover type maps developed to support timber harvest plans and updated forest cover types on the map if a stand was harvested during the study. We classified other cover types as forested wetlands, non-forested wetlands (e.g., open water, peat bog), and non-forested uplands (e.g., agricultural fields, landing strips, gravel pits).

We used a geographic information system (GIS) and the PATCH Analyst 4.2.10 extension for ArcMap 9 (Environmental Systems Research Institute, Inc., Redlands, CA; Rempel et al. 2008) to estimate landscape metrics for each year because timber harvests, storms, and vegetation growth changed landscape characteristics between years. For each of the 180 combinations of transect (i.e., landscape), year, and cover type classification, we estimated mean patch size, number of patches, total area of a cover type, proportion of area for each type cover, and mean patch shape (perimeter: area ratio) within each landscape. In addition, we calculated Shannon's diversity index and Shannon's evenness index of cover types within each landscape using FRAGSTATS Spatial Pattern Analysis program (McGarigal and Marks 1995) in the PATCH extension. We also calculated road density (km/ha) within each landscape by dividing the total length of roads (i.e., surfaces on which on-highway vehicle use is permitted) within an individual landscape by that landscape's total area.

Data Analysis

Model development.—We developed 2 sets of a priori models. Our first model set contained all the models created by Zimmerman et al. (2009) plus a few additional models (see below). The purpose of this first set was to evaluate whether the results of Zimmerman et al. (2009) were applicable over broader areas because their study area had unnaturally high heterogeneity as a result of a long history of forestry experiments. We then designed a second set of models to disentangle the confounding of Shannon's evenness index with cover type dominance reported by Zimmerman et al. (2009). We also examined the relationship

between road density and grouse density because Small et al. (1991) showed that public land hunting areas incurred greater grouse harvest than privately held land. We hypothesized that within our study areas, greater road densities might confer greater hunting exposure than areas with lesser road densities because of differences in access in a manner similar to a public-private land contrast. In addition, neither hunting nor public general access were allowed on the study area used by Zimmerman et al. (2009) so we felt a road density variable might be informative relative to impacts by humans. Zimmerman et al. (2009) examined the relationship between ruffed grouse density and landscape fragmentation, cover type patch size, cover type shape, and diversity of cover types, including Shannon's evenness index, which is a measure of the distribution of area among cover types (Table 1, models 1–9). We also included novel models that partitioned the conifer cover type into pine and spruce-fir cover types, combinations of variables from the top-ranked models in Zimmerman et al. (2009) (e.g., including Shannon's evenness index and mean patch size of aspen and conifer cover types in the same model to reduce the confounding of cover type with the evenness index; Table 1, models 10–18), and several models defining more specific land cover classifications (e.g., mixed hardwood-pine). In addition, we included age class to assess the hypothesis that a mixture of different age classes of aspen was correlated with greater grouse densities (Gullion and Alm 1983). All of these novel models were designed to partition confounding effects and to take advantage of our much larger sample size than Zimmerman et al. (2009) and our selection of 2 study areas that contrasted by dominate cover type (aspen and conifer). We chose these 2 study areas to eliminate the constraints faced by Zimmerman et al. (2009).

In our second model set, we considered 13 additional models (Table 2). To account for cyclic variation in grouse density, we included a year-effect-only model (model 19, Table 2) and included year as a variable in all models (in both model sets), except in the intercept-only model (model 9, Table 1). Zimmerman et al. (2009) suggested that the confounding effect of Shannon's evenness index and cover type might be partitioned by conducting similar research in a landscape dominated by aspen (see above). Because our southern study area was aspen-dominated and cover types within both study areas could be classified by species composition, we were able to model this hypothesis. We considered transect type as a factor (i.e., aspen, conifer, mixed; model 20, Table 2). In addition, we originally classified each study area as either conifer- or aspen-dominated (model 21, Table 2). If conifer cover types had a negative effect on grouse populations and aspen cover types had a positive effect on grouse populations (Gullion and Marshall 1968, Gullion and Alm 1983), we predicted that grouse densities would be lesser in the conifer study area and greater in the aspen study area. We considered road density because we were restricted to using roads or trails when selecting survey transects and roads can have negative effects on wildlife and allow easier access by hunters (Fischer and Keith 1974; models 22, 23, 28–31, Table 2). Hunting was

Table 1. A priori models designed to examine the applicability of existing hypotheses about relationships between density of male ruffed grouse and landscape configuration in northern Minnesota, 2009–2010.

Model no.	Model	K ^a	Variables ^b and predictions ^c
1	Mean patch size: aspen and conifer ^d	5	–Year ± MPS _A – MPS _C
2	Mean patch size: all cover types ^d	8	–Year ± MPS _A – MPS _C – MPS _N – MPS _M ± MPS _W
3	Fragmentation: aspen ^d	5	–Year ± NP _A ± MPS _A
4	Fragmentation: aspen and conifer ^d	7	–Year ± NP _A ± MPS _A – NP _C – MPS _C
5	Mean patch shape: aspen and conifer ^d	5	–Year ± MPA _A + MPAR _C
6	Mean patch size and shape: aspen and conifer ^d	7	–Year ± MPS _A – MPS _C ± MPAR _A + MPAR _C
7	Shannon's diversity index ^d	4	–Year ± SDI _{ACNMW}
8	Shannon's evenness index: all cover types ^d	4	–Year ± SEI _{ACNMW}
9	Intercept only	2	None
10	Mean patch size: aspen and conifer (Zimmerman 2006) ^d and age class ^d	9	–Year ± MPS _{A 0–7} ± MPS _{A 8–25} ± MPS _{A >25} – MPS _{C 0–20} – MPS _{C 21–50} – MPS _{C >50}
11	Mean patch size: aspen, spruce-fir, pine ^e	6	–Year ± MPS _A – MPS _P ± MPS _{SF}
12	Mean patch size and shape: aspen, spruce-fir, pine ^e	9	–Year ± MPS _A – MPS _{SF} – MPS _P ± MPAR _A + MPAR _{SF} – MPAR _P
13	Fragmentation: aspen, spruce-fir, and pine ^e	9	–Year ± NP _A ± MPS _A ± NP _{SF} ± MPS _{SF} – NP _P – MPS _P
14	Shannon's evenness index: all cover types ^e (Zimmerman 2006) and age class ^e	4	–Year ± SEI _{A 0–7, A 8–25, A >25, C, N, M, W}
15	Shannon's evenness index: aspen and conifer ^d	4	–Year ± SEI _{A, C}
16	Shannon's evenness index: all cover types (subdivided) ^e	4	–Year ± SEI _{A, SF, P, N, MSF, MP, FW, NFW, NFU}
17	Shannon's evenness index: all cover types ^d and mean patch size: aspen and conifer ^d	6	–Year ± SEI _{ACNW} ± MPS _A – MPS _C
18	Shannon's evenness index: all cover types ^d and mean patch size and shape: aspen and conifer	8	–Year ± SEI _{ACNW} ± MPS _A – MPS _C ± MPAR _A + MPAR _C

^a Number of parameters.^b Year, year effect (2010 is reference year); NP, number of patches; MPS, mean patch size; MPAR, mean perimeter to area ratio; SDI, Shannon's diversity index; SEI, Shannon's evenness index; TT, transect type. Subscripts represent habitat types: A, aspen; C, conifer; N, northern hardwood; W, wetland; M, mixed hardwood-conifer; MP, mixed hardwood-pine; MSF, mixed hardwood-spruce-fir; SF, spruce fir; P, pine; FW, forested wetland; NFW, non-forested wetland; NFU, non-forested uplands.^c + = predicted positive correlation, – = predicted negative correlation, ± = correlation could be positive or negative.^d Zimmerman et al. (2009) original model or variable classification.^e Zimmerman et al. (2009) combination of original models or finer scale variable classification.

permitted on our study area and vehicle traffic along roads in our study area appeared to be greater during the fall hunting seasons than at other times of the year. Thus, we predicted that the density of roads would be negatively correlated with grouse density. In addition to road-based models, we also predicted that the proportion of the aspen cover type within

landscapes would be positively related to grouse density (models 24, 25, 30–31, Table 2). Neither of these effects (hunting or increased traffic during fall) occurred on the study area used in Zimmerman et al. (2009). Aspen stands >25 years old provide the primary winter food for grouse; thus, we predicted that grouse density would be positively

Table 2. A priori models examining novel relationships between landscape characteristics and male ruffed grouse density in northern Minnesota, 2009–2010.

Model no.	Model	K ^a	Variables ^b and predictions ^c
19	Year only	3	–Year
20	Transect type	4	–Year ± TT _{ACM}
21	Study area	4	–Year – SS _{AC}
22	Road density	4	–Year – RD
23	Road density and Shannon's evenness index ^d	5	–Year – RD ± SEI _{ACNMW}
24	Proportion of aspen ^d	4	–Year ± Prop _A
25	Proportion of aspen and Shannon's evenness index ^d	5	–Year + Prop _A ± SEI _{ACNMW}
26	Mean patch size aspen (mature) ^e	4	–Year ± MPS _{a>25}
27	Mean patch size aspen ^e	4	–Year ± MPS _a
28	Road density and mean patch size aspen ^d	5	–Year ± MPS _A – RD
29	Road density, mean patch size aspen ^d and Shannon's evenness index ^d	6	–Year ± MPS _A – RD ± SEI _{ACNMW}
30	Road density, proportion of aspen ^d , and Shannon's evenness index ^d	6	–Year ± Prop _A – RD ± SEI _{ACNMW}
31	Road density and proportion of aspen ^d	5	–Year – RD ± Prop _A

^a Number of parameters.^b Year, year effect (2010 is reference year); SS, study area; RD, road density; Prop_A, proportion of aspen; MPS, mean patch size; SEI, Shannon's evenness index; TT, transect type. Subscripts represent habitat types: A, aspen; a, aspen subdivided; C, conifer; N, northern hardwood; W, wetland; M, mixed hardwood-conifer.^c + = predicted positive correlation, – = predicted negative correlation, ± = correlation could be positive or negative.^d Zimmerman et al. (2009) variable classification.^e Finer scale classification (i.e., including younger age classes of aspen than Zimmerman et al. 2009).

correlated with mature aspen stands within the landscape (Jakubas and Gullion 1991; model 26, Table 2). That is, landscapes that contain winter food patches should contain more grouse because grouse do not switch from a winter diet until halfway through the breeding season. We predicted that grouse density would be positively related to mean patch size of aspen in all age classes because grouse use aspen of different age classes. We also developed a finer scale classification that identified young aspen patches (Gullion and Alm 1983, Rusch et al. 2000; model 27, Table 2), which could not be included in the study by Zimmerman et al. (2009) because of their small sample size of transects.

Model selection and estimation of parameters.—We used information-theoretic model selection to assess the relative fit of models to our data (Burnham and Anderson 2002). We ranked models using Akaike's Information Criterion for small samples (AIC_c ; Burnham and Anderson 2002). We calculated AIC_c values using the "qpcR" package (Ritz and Spiess 2008) in program R (R Development Core Team 2010). Our response variable was grouse density (the number of observed unique male grouse within a landscape divided by the total area [ha] of that landscape). We chose to model counts of grouse using density as a response for 3 reasons. First, density estimates are continuous variables; thus, analysis would be based on a normal distribution, whereas an alternative use of our count data could be based on a discrete distribution (negative binomial, Poisson). Normal-based analyses are more robust to departures from normality, for example, than Poisson-based models are to departures from a Poisson distribution, although this issue can be dealt with adequately (White and Bennetts 1996). Second, density is a standard exposition in the wildlife literature of wildlife abundance relative to area of interest. Third, using count data in Poisson-based analyses might mislead biologists into using drumming count data collected on annual spring surveys as equivalent to our count data based on a robust design (Zimmerman and Gutiérrez 2007). For transects in which a covariate was undefined (e.g., a mean patch size cannot be calculated for a cover type not present), we used the mean value for that variable among all other transects. We feel this was acceptable for several reasons. First, the data were missing at random. The data were actually undefined because of lack of certain cover types in certain landscapes, not because of potentially nonrandom influences on missing data. Second, we were not attempting to impute values for missing data from existing observations; rather, we were simply using these mean values as place holders so we could fit the models without excluding the affected landscapes from the analysis or biasing the parameter estimates. However, it did result in underestimation of the standard error of the affected parameter estimates because the sample size for those variables was artificially large.

We used mixed effects models in a linear regression framework to estimate the parameters of models relating grouse density to landscape characteristics. We used the lmer function from the "lme4" package (Pinheiro et al. 2011) in program R (R Development Core Team 2010) to fit our

models. We treated individual transects as a random effect and all other variables as fixed effects. The general structure of these models was:

$$D_{ij} = b_i + \sum_k x_{ijk}\beta_k + \varepsilon_{ij}$$

where D_{ij} was our response variable (grouse density) for transect i of year j , b_i was the transect-specific intercept, x_{ijk} were the covariate data for the k explanatory variables, β_k were the slope coefficients, and ε_{ij} was the error term.

After fitting models to our data, we used diagnostic plots to assess the assumptions of normality and constant variance of residuals. Three landscapes that had high grouse densities contributed to apparent heteroscedasticity of the residuals. To mitigate this violation of model assumptions, we applied transformations to the response variable (grouse density) and the predictor variables (e.g., Shannon's evenness index) in an attempt to normalize the residuals, and thus reduce the non-constant variance of the residuals. However, no transformation improved the normality and non-constant variance of the residuals; the residual pattern was likely caused by 3 observations of landscapes having high grouse density, which we determined to be valid based on our sampling protocol. Therefore, we fitted our models using ordinary least squares regression. We then conducted a weighted least squares regression (Neter et al. 1996:400–409) to reduce the influence of these observations.

RESULTS

Regression coefficients estimated using weighted least squares methods did not differ substantially from the regression coefficients estimated using ordinary least squares. Applying weighted least squares methods did not improve the non-constant variance resulting from 3 extreme values of grouse density. Therefore, we based inference on results from ordinary least squares.

We conducted 480 grouse surveys on our 30 landscapes during our 2 years of study (2009, 2010). We located 432 and 449 grouse drumming structures during 2009 and 2010, respectively. Of these structures, we estimated 290 (2009) and 230 (2010) were primary drumming structures, which represented individual grouse activity centers (Rusch et al. 2000). Densities of grouse ranged from 0.013 grouse/ha to 0.151 grouse/ha among landscapes in 2009 and from 0.012 grouse/ha to 0.143 grouse/ha among landscapes in 2010. Grouse density was greater during 2009 (0.07 grouse/ha, SD = 0.03) than in 2010 (0.05 grouse/ha, SD = 0.03; model 9 $\beta_{\text{year2010}} = -0.014$).

The intercept-only (null) model was the highest ranked model in our first model set (model 9; $AIC_c = -250.680$; see also expanded model set 1 results in Appendix S2, available online at www.onlinelibrary.wiley.com). We identified 1 competing model and it was consistent with Zimmerman et al. (2009), suggesting that landscapes having a more even distribution of cover types had greater grouse densities (model 8; $AIC_c = -249.740$, $\beta_{\text{evenness}} = 0.090$, 95% CI = 0.017–0.163, Fig. 1; see also Table S1 for expanded model

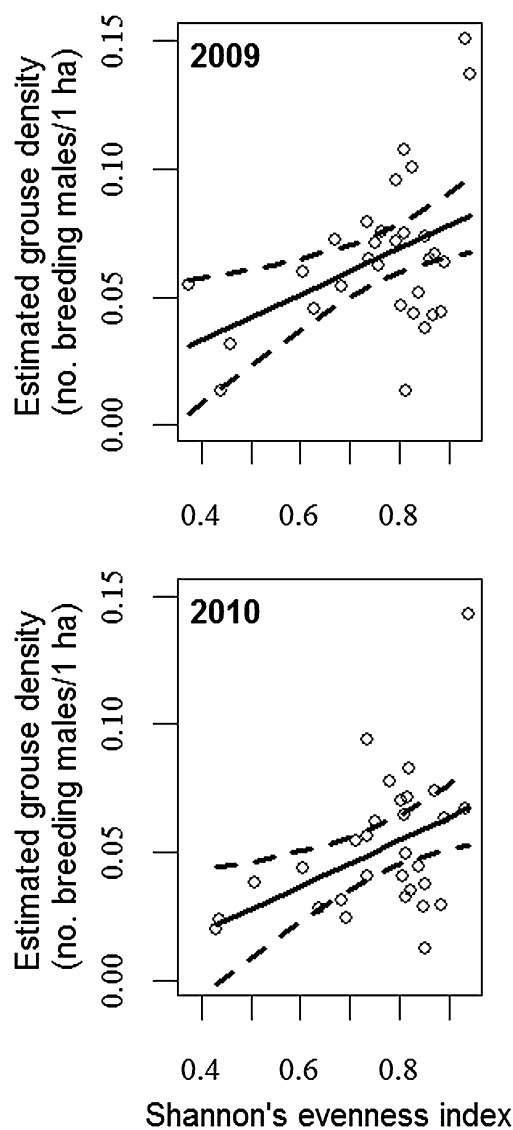


Figure 1. Estimated relationship between Shannon's evenness index and male ruffed grouse density in northern Minnesota during 2009 and 2010. Dashed lines represent the 95% confidence interval; points represent observed data.

set 1 results in Appendix S2, available online at www.onlinelibrary.wiley.com). However, only a modest amount of variation in grouse density was explained by this model (coefficient of determination, $R^2 = 0.18$).

A model with Shannon's evenness index based on a more detailed classification of forest cover types (model 14; $AIC_c = -248.052$; see also Table S1 for expanded model set 1 results in Appendix S2, available online at www.onlinelibrary.wiley.com) was <2 AIC_c units from the model with a Shannon's evenness index based on a more simplified classification of forest cover types (i.e., without including aspen age classes, model 8). However, both models had the same number of parameters, so the inclusion of aspen age class in the cover type classification did not improve the correlation between Shannon's evenness index and grouse density (model 14, $\beta_{\text{evenness aspen age}} = 0.058$, 95% CI = 0.006–0.110).

In our second model set, a model based on road density was the top model predicting grouse density within landscapes and ranked higher than all other models in both sets (model 22, Table 3; see also Table S2 for expanded model set 2 results in Appendix S2, available online at www.onlinelibrary.wiley.com). Although the confidence limits of the parameter estimate slightly overlapped zero and the $R^2 = 0.12$, this model indicated that the greater the density of roads within landscapes, the lesser the grouse density ($\beta_{\text{road density}} = -0.860$, 95% CI = -1.76 – 0.04 ; Table 3, Fig. 2).

The model containing Shannon's evenness index and year (model 8, Table 3) was ranked higher than models containing the proportion of aspen within a landscape and year (model 24, $\beta_{\text{proportion aspen}} = 0.066$, 95% CI = 0.126 – 0.006 ; Table 3; see also Table S3 for expanded set of combined model sets 1 and 2 comparison in Appendix S2, available online at www.onlinelibrary.wiley.com). This result supported the hypothesis that Shannon's evenness index was a better indicator of grouse density than the proportion of the aspen cover type within our landscapes. In addition, the mean patch size of aspen was negatively correlated with the mean patch size of conifer (Pearson correlation coefficient

Table 3. Model selection results combining top models from two model sets (1 and 2) containing 95% of Akaike weights (w_i) for assessing the relationships between landscape characteristics and male ruffed grouse density in northern Minnesota, 2009–2010.

Model	Set		ΔAIC_c^a	K^b	AIC_c^c	w_i^d
22	2	Road density	0	4	–252.21	0.501
9	1	Intercept only ^e	1.531	2	–250.68	0.233
23	2	Road density and evenness	2.324	5	–249.89	0.157
8	1	Shannon's evenness index: all cover types ^e	2.472	4	–249.74	0.146
19	2	Year only	2.571	3	–249.64	0.138
31	2	Road density and proportion of aspen	3.048	5	–249.16	0.109
24	2	Proportion of aspen	4.139	4	–248.07	0.063
14	1	Shannon's evenness index: all cover types ^e and age class	4.160	4	–248.05	0.063

^a Difference in Akaike's Information Criterion corrected for small sample size (AIC_c) compared to the best-fitting model.

^b Number of parameters in the model.

^c Small sample Akaike's Information Criterion.

^d Akaike's weights.

^e Zimmerman et al. (2009) original model or variable classification.

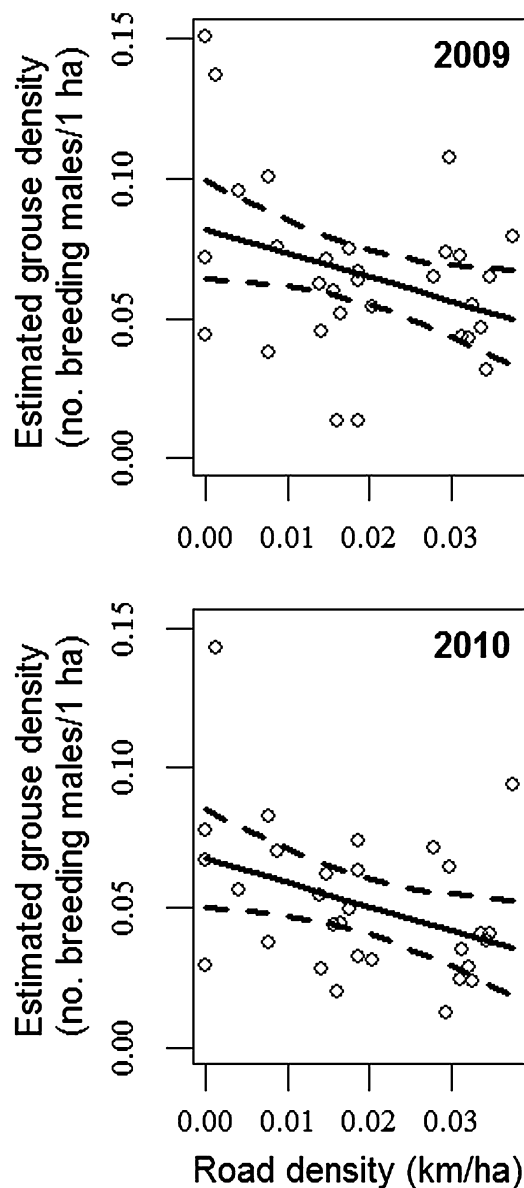


Figure 2. Estimated relationship between road density and male ruffed grouse density in northern Minnesota during 2009 and 2010. The dashed lines represent the 95% confidence interval; points represent observed data.

$= -0.36, P \leq 0.001$). More importantly, the proportions of aspen and conifer within landscapes were correlated with Shannon's evenness index, in the same manner as reported by Zimmerman et al. (2009). That is, landscapes having greater densities of grouse and more evenly distributed cover types also had a lesser proportion of conifer (Pearson correlation coefficient, $r = -0.79, P < 0.001$; Fig. 3) and a greater proportion of aspen (Pearson correlation coefficient, $r = 0.55, P < 0.001$; Fig. 3). The model based on the year effect alone suggested that grouse density declined from 2009 to 2010 (model 19, $\beta_{\text{year2010}} = -0.014$, 95% CI = -0.024 to -0.005).

DISCUSSION

Wildlife managers since the time of Aldo Leopold (1933) have been interested in the relationship between landscape

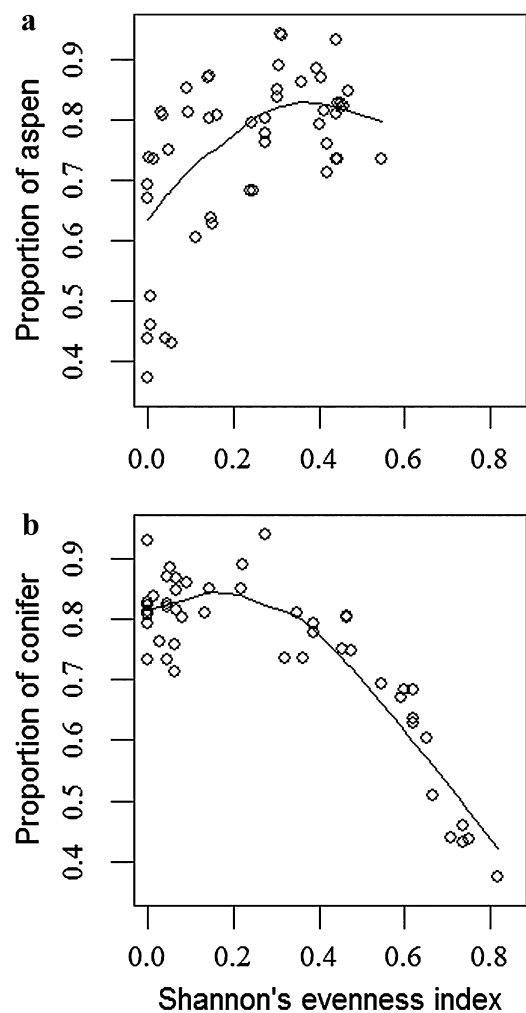


Figure 3. Observed relationship between Shannon's evenness index and the proportion of the (a) aspen and (b) conifer cover types within the landscape in northern Minnesota during 2009 and 2010.

cover types (composition) and their arrangement (inter-spersion) and a species' density. Gullion (1988) specifically recommended the interspersed of different age classes of aspen in close proximity to provide such a favorable juxtaposition for ruffed grouse. However, grouse often are found in landscapes that have mixtures of cover types that may or may not include aspen (Zimmerman et al. 2009, Stauffer 2011). Zimmerman et al. (2009) found that Shannon's index was correlated with greater ruffed grouse density. However, their study was conducted in an experimental forest where landscapes were highly heterogeneous because of a history of forest experiments. Thus, their results may not be applicable to the majority of grouse range because their study area was managed differently than most other areas. Our study demonstrated that the use of Shannon's index was correlated with grouse density over large areas of publically managed land. Although our top model was an intercept-only model, this model likely reflected the broad use by grouse of many cover types within landscapes when their population was at peak

abundance, which would be predicted by habitat selection theory (Fretwell and Lucas 1970, Fretwell 1972).

Although we expanded the applicability of the findings from Zimmerman et al. (2009) to larger areas, and perhaps the Laurentian Floristic Province, we did not resolve the confounding of evenness (i.e., the distribution of area among cover types) and cover type (aspen and conifer) even though we designed our study specifically to resolve this confounding relationship. This suggested that either unknown ecological relationships occur between cover type and evenness or landscapes managed to produce conifers for timber production lacked the spatial heterogeneity or food necessary to support high grouse densities. For example, conifer species, such as red pine, were typically managed as monocultures in our study area. In contrast, aspen regenerated naturally after harvest, which resulted in greater spatial heterogeneity. Shannon's evenness index was a better predictor of grouse density at the landscape scale than cover type dominance, which supported our suggestion that conifer plantation management resulted in less diverse landscapes than natural landscapes. Other models representing hypotheses about relationships with grouse density such as 1) the mean size of aspen patches and 2) the proportions of aspen and conifer within the landscape had less support than the Shannon's evenness index model, which again supported our inference that Shannon's evenness index was a better predictor of grouse density.

Zimmerman et al. (2009) suggested that the correlation between Shannon's evenness index and grouse densities within the landscape may have been related to variation in habitat needed by grouse during different life history stages (e.g., male drumming vs. female nesting). Male drumming locations have been shown in many studies to be in dense forest stands or thicket-like shrub cover, whereas nests tend to be located in much less dense stands (Gullion and Marshall 1968, Gullion 1988, Rusch et al. 2000). Zimmerman et al. (2009) also hypothesized that grouse may have selected landscapes having an even distribution of cover types because such landscapes allow grouse to access different cover types during inclement weather (Thompson and Fritzell 1988).

We found that grouse density was greater in landscapes having lesser road density. Our observations were in contrast to those made in the Appalachian region where grouse selected areas having greater road densities (Tirpak et al. 2010). However, Tirpak et al. (2010) suggested that vegetation structure found along roads (i.e., disturbed habitats resembling early successional stages) likely provided the only grouse habitat within their study area. Early successional habitats were likely not limiting on our study area because aspen was a dominant cover type (58.2% of upland habitat) and 44.5% of aspen stands were ≤ 20 years old. Alternatively, the relationship between road density and grouse density was confounded by cover types and evenness. The majority of roads occurred along ridges (uplands) that could support road construction more easily than surrounding lowlands; these areas also supported more intensive conifer management (e.g., 30% of cover within 200 m of a

road was conifer, whereas it comprised only 13% over the entire study area), which reduced evenness and created areas unsuitable for grouse along roads.

Fischer and Keith (1974) reported that hunting mortality of adult territorial males increased with decreased distance to roads. Although many studies have concluded that mortality from sport hunting does not adversely affect populations of grouse; DeStefano and Rusch (1986) noted that such studies were conducted within large areas of contiguous habitat. They hypothesized that immigration offset local effects of hunting. Small et al. (1991) suggested that high hunting mortality could decrease ruffed grouse numbers in fragmented habitats when immigration could not offset hunting mortality. We did not expect any limitation of grouse dispersal, so lesser grouse densities in landscapes with greater road densities likely reflected habitat conditions (i.e., more conifer plantations) than effects of hunting.

Observers possibly took less time to survey transects that had a greater road density (i.e., improved roads are easier to traverse than trails) such that a lesser estimated density reflected lesser survey effort. However, survey duration was a function of the number of displaying males detected during a survey and the transect's total length. Mean survey time (minutes) per unit area of landscape (ha) was correlated with the average number of grouse in a landscape (Pearson's correlation coefficient = 0.7, 95% CI = 0.456–0.847) and not road density (Pearson's correlation coefficient = -0.32, 95% CI = -0.608–0.048). Survey routes with many displaying males took longer to survey regardless of the road density because we had to locate males visually and to find their drumming stages.

Despite the correlation of the Shannon's evenness index and road density variables with grouse density, the models that we considered explained only a small proportion of variation in grouse density within landscapes. We hypothesized this was because peak abundance of grouse during their 10-year cycle masked the effects of selection for optimal habitats or landscapes (Fretwell and Lucas 1970). In addition, even though we designed our study to disentangle the effects of cover type and evenness reported by Zimmerman et al. (2009), we were unable to do so. This is an important question because it has implication for the forest management effects on grouse. Thus, we suggest that this confounding can best be resolved by experiments that manipulate cover types and their distribution in ways that will lead to strong inference on the relative roles of cover type evenness and the types of cover (aspen or conifer) on grouse density. When conducting experiments or habitat selection studies on a species that undergoes population cycles, one must consider the stage of the population cycle because habitat selection is influenced not only by the presence of resources, but also by other individuals. Moreover, it suggests that the broadest inferences should be based on results from studies conducted that encompass all phases of the cycle. The stage of a cycle also allows multiple predictions about population responses because of varying density that can be used to gauge results from experiments (Zimmerman et al. 2009).

MANAGEMENT IMPLICATIONS

Grouse respond positively to habitat modification, particularly where aspen mosaics are favored (Gullion and Alm 1983, Rusch et al. 2000). We recommend Shannon's evenness index be used as a tool to plan long-term interspersions of available cover types in the Laurentian Mixed Forest Province of North America if ruffed grouse are a management objective. High evenness reflects cover type composition and distribution on a landscape that is correlated with high grouse densities. In contrast, landscapes dominated by conifer plantations, which have low evenness indices, do not contain high densities of grouse. Thus, if both conifers and grouse are desired management objectives, then managers should either emphasize regeneration strategies that result in stands having either mixed-species composition or intersperse conifer plantations with aspen stands. Moreover, we suggest that management should occur not only at the scale of individual forest stands but also at the scale of landscapes to foster creation of mosaics of more evenly distributed cover types over larger areas.

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Supporting Information

Additional supporting information may be found in the online version of this article at the publisher's web-site.

Appendix S1. Supplemental methods.

Figure S1. Map of study area in Minnesota, USA.

Table S1. Criteria applied in a sequential manner during 2009 to distinguish primary drumming structures from secondary drumming structures on the Red Lake Wildlife Management Area, Roosevelt, MN.

Tables S2. Sequential criteria used in 2010 to distinguish primary drumming structures from secondary drumming structures on the Red Lake Wildlife Management Area, Roosevelt, MN. The essential difference between the two sets of criteria is the use of field notes by observers. Observers

were trained to record specific information based on an assessment of 2009 criteria and field work.

Appendix S2. Additional results.

Table S1. Model selection results for first model set (only those models with $w_i > 0.001$) for assessing the relationships between landscape characteristics and male ruffed grouse density in northern Minnesota during 2009–2010.

Table S2. Model selection results for second model set (only those models with $w_i > 0.001$) for assessing the relationships between landscape characteristics and male ruffed grouse density in northern Minnesota during 2009–2010.

Table S3. Model selection results for all models (models with $w_i > 0.001$) for assessing the relationships between landscape characteristics and male ruffed grouse density in northern Minnesota during 2009–2010.